

JIATONG WEI

TEL: (+86) 181-6191-0732 | weijiatong@mail.nwpu.edu.cn | jiatong-wei.github.io/profile/

EDUCATION

Northwestern Polytechnical University, China (NPU)

09/2023–06/2027 (Expected)

B.Eng. in Underwater Acoustic Engineering, School of Marine Science and Technology

- Average grade: 85/100; comprehensive evaluation ranking: top 5%; awarded Outstanding Undergraduate and University Scholarship.
- Selected honors include National Grand Prize in the National Marine Vehicle Design and Production Competition and two Provincial First Prizes in the China Undergraduate Engineering Practice and Innovation Ability Competition.
- English: CET-4 571; CET-6 passed.

RESEARCH PROFILE

Interests: Robotics, Physical AI, underwater unmanned systems, embodied intelligence, and full-stack robotic system development.

Engineering Foundation: Hands-on experience across mechanical modeling, embedded control, robot chassis design, simulation, and prototype integration.

Working Style: Able to communicate in English for technical collaboration; regularly reads English academic papers and follows robotics / Physical AI deployment.

RESEARCH OUTPUTS AND INTELLECTUAL PROPERTY

- One design patent accepted.
- Two computer software copyrights authorized as a core member.
- Independent 3DGS content ranked among the top 100 by views on a relevant content platform; work used for NPU aircraft-display content.

RESEARCH EXPERIENCE

Intelligent Transportation Platform Based on Omnidirectional Wheels and Deep Learning

05/2024–04/2025

Undergraduate Innovation and Entrepreneurship Training Program | Sole Lead

- Designed a Mecanum-wheel chassis and developed the STM32F407-based low-level controller, connecting mechanical design, embedded control, and upper-level autonomy into one robotic prototype.
- Deployed SLAM algorithms on RK3588, built a behavior-cloning data collection framework in Isaac Sim, implemented Sim-to-Real end-to-end visual obstacle avoidance, colorized point clouds with RGB images, and integrated YOLOv5 for low-latency person following.
- Delivered a prototype with autonomous mapping, navigation, and gamepad remote-control capabilities, supporting the successful completion of the project.

SELECTED ROBOTICS AND ENGINEERING PROJECTS

Integrated Underwater Exploration and Intervention Platform

08/2025

National Grand Prize, National Marine Vehicle Design and Production Competition | Algorithm and Prototype Team Member

- Proposed a DQN-based underwater manipulator trajectory planning algorithm for efficient and stable trajectory control in complex underwater environments.
- Assisted in solving cross-media water-air communication issues during prototype development, enabling underwater wireless control without power-line carrier communication.
- Participated in pool testing to verify overall structural strength and the effectiveness of motion-control algorithms.

Smart Logistics Handling Robot

04/2025

Provincial First Prize, China Undergraduate Engineering Practice and Innovation Ability Competition | Embedded Control Contributor

- Designed a cascaded and coupled speed-position-angle PID controller, upgrading the original open-loop scheme to closed-loop GMR motor control and reducing accumulated angular drift during robot motion to within $\pm 0.1^\circ$.
- Implemented Mecanum-wheel chassis kinematics and attitude solving on an STM32F103 MCU, and used LoRa communication for wireless chassis parameter debugging.
- Worked with teammates on joint debugging, improved target-hitting accuracy to 90%, and supported one-click launch through a serial touch screen.

Multifunctional Revolving-Body Marine Vehicle

07/2025

Northwest Regional First Prize, National Marine Vehicle Design and Production Competition | Independent System Designer

- Independently completed the overall system design and physical prototype development, gaining hands-on familiarity with watertight cabin sealing and FDM structural reinforcement.
- Designed two vector-propulsion mechanisms and improved a TVC vector-thrust structure into a dual-helical-gear vector-propulsion mechanism, addressing motor heat dissipation and cabin intrusion challenges.
- Planned module functions and clarified key technical links in perception, planning, and control for underwater unmanned systems.
- Built a practical understanding of the path from system-level design to engineering implementation for underwater autonomous unmanned systems.

Smart Agriculture Digital Twin Workshop

11/2025

National First Prize, National 3D Digital Innovation Design Competition | Unreal Engine and HIL Developer

- Developed a complete smart-agriculture digital twin system from scratch in Unreal Engine, enabling users to experience an immersive digitalized agriculture workflow.
- Developed the quiz system and real-time UAV irrigation function; built a hardware-in-the-loop simulation environment; implemented UAV flight-control logic with Blueprint; and connected virtual irrigation tasks with the physical quiz system through data interaction.

HONORS AND AWARDS

National Grand Prize – National Marine Vehicle Design and Production Competition	08/2025
Northwest Regional First Prize – National Marine Vehicle Design and Production Competition	07/2025
Provincial First Prize twice – China Undergraduate Engineering Practice and Innovation Ability Competition	2025
National First Prize – National 3D Digital Innovation Design Competition	11/2025
2026 Mathematical Contest in Modeling	Meritorious Winner
13th Future Designer National College Digital Art and Design Awards, National Finals	National Third Prize

TECHNICAL SKILLS

Programming and Embedded: Python, C, STM32F407/F103 development, low-level control, chassis kinematics, PID control, LoRa-based debugging.

Robotics and Simulation: ROS2, Isaac Sim, SLAM deployment, behavior-cloning data collection, Sim-to-Real visual obstacle avoidance, YOLOv5 integration.

UAV Flight-Control Practice: Two consecutive years of participation in the UAV track of the National Undergraduate Electronics Design Contest; experienced in PX4 and ANO flight-controller tuning.

Design and Tools: SolidWorks, Keil, Git, Linux, Unreal Engine, Blueprint, mechanical modeling, PCB design, hardware driver development.

Agentic Engineering: Practical experience using Codex and mainstream CLI-based toolchains for engineering workflows.